Resultater fra QH4 for Hilde Celine Løvland

Resultat for dette forsøket: 11 av 11

Innlevert 1. okt. i 15.29

Dette forsøket tok 14 minutter

Riktig svar

Spørsmål 1

5 / 5 poeng

Hydraulic system . Consider a variable displacement pump set at a constant pressure supply and connected to a base unit, see Figure Q4.8 . The base unit is a 4/3-way proportional DCV and an extracting cylinder, see Figure Q4.3 . Pump max displacement (DMax) = 12 cm^3/rev . Pump speed (n) = 1470 rev/min . Pump volumetric efficiency (eta_vP) = 0.92 . Main crack pressure (p_crM) = 230 bar . The 4/3-way proportional directional control valve has identical discharge coefficients: C_dPA = C_dBT = C_dPB = C_dAT = C_d . The 4/3-way proportional directional control valve has symmetrical discharge areas: A_dMax,PA = A_dMax,PB and A_dMax,BT = A_dMax,AT . Liquid density (rho) = 875 kg/m^3 . Discharge coefficient (C_d) = 0.70 . Maximum discharge area (A_dMax,PA) = 16 mm^2 . Maximum discharge area (A_dMax,BT) = 17 mm^2 . Max valve spool travel (x_Max) = 10.0 mm . Valve spool travel (x) = 5.4 mm . The cylinder is extracting . The cylinder is subjected to a resistive (positive) load . Cylinder diameter (d) = 63 mm . Cylinder rod diameter (dR) = 40 mm . Cylinder hydromechanical efficiency (eta_hmC) = 0.92 . Cylinder load force (F_L) = 54.3 kN . Pump set pressure (p_set) = 195 bar . What is the pressure (p_A) in bar?

190,2366

190,2366 (med marginen: 1,9024)

Riktig svar

Spørsmål 2

1 / 1 poeng

Valve bandwidth requirement . Consider a motor connected to a servo valve, see Figure Q4.10 . The two hydraulic lines are identical, i.e., V_L1 = V_L2 and V_L = V_L1 + V_L2 . Bulk modulus (beta) = 1254 MPa . Motor displacement (D) = 710 cm^3/rev . Total line volume (V_L) = 0.22 l . Mass moment of inertia (J) = 26.98 kg*m^2 . What is the minimum reuired valve bandwidth (f_v@90deg) in Hz?

24,1233

24,1233 (med marginen: 0,2412)

Riktig svar

Spørsmål 3

2 / 2 poeng

Minimum rated flow for servo valve . Consider a motor connected to a servo valve, see Figure Q4.12 . Motor displacement (D) = 32 cm^3/rev . Mass moment of inertia (J) = 0.31 kg*m^2 . Supply pressure (p_s) = 250 bar . Payload total rotation (Delta-theta) = 79.6 rad . Total task time (Delta-t) = 2.497 s . Ramp time (t_R) = 0.617 s . External moment (M_1) = -48 . External moment (M_2) = 1 . External moment (M_3) = 14 . Consider a specific time during the task (t) = 0.712 s . What is the no-load flow (Q_NL) at the considered time (t) in l/min?

12,9894

12,9894 (med marginen: 0,1299)

Riktig svar

Spørsmål 4

3 / 3 poeng

No-load flow in servo systems . Consider a motor connected to a servo valve, see Figure Q4.12 . Motor displacement (D) = 355 cm^3/rev . Mass moment of inertia (J) = 9.63 kg*m^2 . Supply pressure (p_s) = 250 bar . Valve rated pressure (p_r) = 70 bar . Payload total rotation (Delta-theta) = 81.8 rad . Total task time (Delta-t) = 6.149 s . Ramp time (t_R) = 1.703 s . External moment (M_1) = 453 . External moment (M_2) = -217 . External moment (M_3) = -19 . What is the minimum required valve flow (Q_r,min) in l/min?

42,4089

42,4089 (med marginen: 0,4241)

Quizresultat: 11 av 11